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Department of Computer Science and Engineering (CSE)

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Aashnan Rahman, Jr. Lecturer, CSE

Lecture 4

Simplifying Boolean Expressions: Quine McCluskey Method

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1 Quine McCluskey Method

The Prime Implicant method (often used through the Quine–McCluskey procedure) is used to systematically minimize Boolean functions. Instead of relying on visual techniques like K-maps, which become difficult and error-prone for more than four or five variables, this method provides an algorithmic and structured approach that works for any number of variables.

We use it because it guarantees finding all possible simplifications and ensures that the final expression is minimal in terms of Sum of Products (SOP) or Product of Sums (POS). By organizing minterms into groups, combining those that differ by only one bit, and then identifying essential prime implicants through a coverage table, the method removes redundancy in a logical and repeatable way.

In short, it replaces guesswork with a step-by-step procedure, making Boolean simplification scalable, accurate, and suitable for larger digital logic problems where manual grouping becomes impractical. The rules for simplifying Boolean expressions using the Quine McCluskey method with 2 to 6 variables and don't-care conditions.

- Step 0:** Write down the binary representation of the minterms and count the number of 1s in each binary representation.
- Step 1:** Group the minterms based on the number of 1s. (*Preprocessing step*)
- Step 2:** Compare minterms in adjacent groups and combine those that differ by **only one bit**. Mark combined minterms as “used” with a tick.
- Step 3:** Repeat Step 2 until no further combinations are possible. The remaining unmarked minterms are the **prime implicants**.
- Step 4:** Construct a table where Prime Implicants are listed along rows and minterms along columns. For each Prime Implicant, place an “x” under every minterm it covers. If a column contains a single “x”, circle it. Prime Implicants containing at least one circled “x” are the **Essential Prime Implicants**.
- Step 5:** Select all essential prime implicants. The final solution is the **sum of essential prime implicants**.

To be Noted: For expressions containing don't care terms, there should be no change until Step 3. In Step 4, we list the Prime Implicants, and for columns we include only the minterms that are 1. We do not include don't care terms, as our goal is not to cover them. They are included only if needed for simplification.

1.1 Grouping Examples:

Example 1: Valid grouping (1-bit difference)

$$m_5 = 0101 \quad \text{and} \quad m_7 = 0111$$

Comparison: 0101 vs 0111

$$\Rightarrow 01_1$$

(valid grouping $\rightarrow$ only one bit differs)

Example 2: Valid grouping (K-map style simplification)

$$m_2 = 0010 \quad \text{and} \quad m_3 = 0011$$

Comparison: 00**1**0 vs 00**1**1

$$\Rightarrow 001_$$

(valid grouping $\rightarrow$ last bit changes only)

Example 3: Invalid grouping (more than 1-bit difference)

$$m_4 = 0100 \quad \text{and} \quad m_{11} = 1011$$

Comparison: 0**1**00 vs 1**0**11

$\Rightarrow$ NOT valid (differs in 4 bits)

1.2 Worked Out Example

1.2.1 Example 1

$$F(A,B,C,D) = m(0,1,3,7,8,9,11,15)$$

Step 0:

Group	Minterm	Binary Representation			
		A	B	C	D
m_0	0	0	0	0	0
m_1	0	0	0	1	1
m_3	0	0	1	1	2
m_7	0	1	1	1	3
m_8	1	0	0	0	1
m_9	1	0	0	1	2
m_{11}	1	0	1	1	3
m_{15}	1	1	1	1	4

Step 1:

Group	Minterm	Binary Representation				
		A	B	C	D	
0	m_0	0	0	0	0	✓
1	m_1	0	0	0	1	✓
1	m_8	1	0	0	0	✓
2	m_3	0	0	1	1	✓
2	m_9	1	0	0	1	✓
3	m_7	0	1	1	1	✓
3	m_{11}	1	0	1	1	✓
4	m_{15}	1	1	1	1	✓

Step 2:

Group	Matched Pairs	Binary Representation				
		A	B	C	D	
0	m_0, m_1	0	0	0	-	✓
0	m_0, m_8	-	0	0	0	✓
1	m_1, m_3	0	0	-	1	✓
1	m_1, m_9	-	0	0	1	✓
1	m_8, m_9	1	0	0	-	✓
2	m_3, m_7	0	-	1	1	✓
2	m_3, m_{11}	-	0	1	1	✓
2	m_9, m_{11}	1	0	-	1	✓
3	m_7, m_{15}	-	1	1	1	✓
3	m_{11}, m_{15}	1	-	1	1	✓

Step 3:

Group	Matched Pairs	Binary Representation				PI
		A	B	C	D	
0	m_0, m_1, m_7, m_9	-	0	0	-	$\bar{B}\bar{C}$
0	m_0, m_8, m_1, m_9	-	0	0	-	$\bar{B}\bar{C}$
1	m_1, m_3, m_9, m_{11}	-	0	-	1	$\bar{B}D$
1	m_1, m_9, m_3, m_{11}	-	0	-	1	$\bar{B}D$
2	m_3, m_7, m_{11}, m_{15}	-	-	1	1	BD
2	m_3, m_{11}, m_7, m_{15}	-	-	1	1	BD

Step 4:

PI	Minterms	0	1	3	7	8	9	11	15	
$\bar{B}\bar{C}$	m_0, m_1, m_8, m_9	($\otimes$)	$\times$			($\otimes$)	$\times$			✓
$\bar{B}D$	m_1, m_3, m_9, m_{11}		$\times$	$\times$			$\times$	$\times$		
CD	m_3, m_7, m_{11}, m_{15}			$\times$	($\otimes$)			$\times$	($\otimes$)	✓

$$F = \bar{B}\bar{C} + CD$$

1.2.2 Example 2

$$F(A, B, C, D) = \sum m(0, 1, 2, 8, 10, 11, 14, 15)$$

Step 0:

Group	Minterm	Binary Representation			
		A	B	C	D
0	m_0	0	0	0	0
1	m_1	0	0	0	1
1	m_2	0	0	1	0
1	m_8	1	0	0	0
2	m_{10}	1	0	1	0
3	m_{11}	1	0	1	1
3	m_{14}	1	1	1	0
4	m_{15}	1	1	1	1

Step 1:

Group	Minterm	Binary Representation				
		A	B	C	D	
0	m_0	0	0	0	0	✓
1	m_1	0	0	0	1	✓
1	m_2	0	0	1	0	✓
1	m_8	1	0	0	0	✓
2	m_{10}	1	0	1	0	✓
3	m_{11}	1	0	1	1	✓
3	m_{14}	1	1	1	0	✓
4	m_{15}	1	1	1	1	✓

Step 2:

Group	Matched Pairs	Binary Representation				
		A	B	C	D	
0	m_0, m_1	0	0	0	—	×
0	m_0, m_2	0	0	—	0	✓
0	m_0, m_8	—	0	0	0	✓
1	m_2, m_{10}	—	0	1	0	✓
1	m_8, m_{10}	1	0	—	0	✓
2	m_{10}, m_{11}	1	0	1	—	✓
2	m_{10}, m_{14}	1	—	1	0	✓
3	m_{11}, m_{15}	1	—	1	1	✓
3	m_{14}, m_{15}	1	1	1	—	✓

Step 3:

Group	Matched Pairs	Binary Representation				PI
		A	B	C	D	
0	m_0, m_2, m_8, m_{10}	—	0	—	0	$\overline{B}\overline{D}$
2	$m_{10}, m_{11}, m_{14}, m_{15}$	1	—	1	—	AC
2	m_0, m_1	0	0	0	—	$\overline{A}\overline{B}\overline{C}$

Step 4:

PI	Minterms	0	1	2	8	10	11	14	15	
$\overline{A}\overline{B}\overline{C}$	m_0, m_1	×	⊗							✓
$\overline{B}\overline{D}$	m_0, m_2, m_8, m_{10}	×		⊗	⊗	×				✓
AC	$m_{10}, m_{11}, m_{14}, m_{15}$					×	⊗	⊗	⊗	✓

$$F = \overline{A}\overline{B}\overline{C} + \overline{B}\overline{D} + AC$$

1.2.3 Example 3

$$F(A, B, C, D) = \sum m(0, 3, 5, 6, 7, 10, 12, 13), \quad d(2, 9, 15)$$

(In the following, don't-care minterms are included to form larger groups but are not required to be covered.)

Step 0:

Group	Minterm	Binary Representation			
		A	B	C	D
0	m_0	0	0	0	0
1	m_2	0	0	1	0
2	m_3	0	0	1	1
2	m_5	0	1	0	1
2	m_6	0	1	1	0
2	m_9	1	0	0	1
2	m_{10}	1	0	1	0
2	m_{12}	1	1	0	0
3	m_7	0	1	1	1
3	m_{13}	1	1	0	1
4	m_{15}	1	1	1	1

Step 1:

Group	Minterm	Binary Representation				
		A	B	C	D	
0	m_0	0	0	0	0	✓
1	m_2	0	0	1	0	✓
2	m_3	0	0	1	1	✓
2	m_5	0	1	0	1	✓
2	m_6	0	1	1	0	✓
2	m_9	1	0	0	1	✓
2	m_{10}	1	0	1	0	✓
2	m_{12}	1	1	0	0	✓
3	m_7	0	1	1	1	✓
3	m_{13}	1	1	0	1	✓
4	m_{15}	1	1	1	1	✓

Step 2:

Group	Matched Pairs	Binary Representation				
		A	B	C	D	
0	m_0, m_2	0	0	—	0	×
1	m_2, m_3	0	0	1	—	✓
1	m_2, m_6	0	—	1	0	✓
1	m_2, m_{10}	—	0	1	0	×
2	m_3, m_7	0	—	1	1	✓
2	m_5, m_7	0	1	—	1	✓
2	m_5, m_{13}	—	1	0	1	✓
2	m_6, m_7	0	1	1	—	✓
2	m_9, m_{13}	1	—	0	1	×
2	m_{12}, m_{13}	1	1	0	—	×
3	m_7, m_{15}	—	1	1	1	✓
3	m_{13}, m_{15}	1	1	—	1	✓

Step 3:

Group	Matched Pairs	Binary Representation				PI
		A	B	C	D	
1	m_2, m_3, m_6, m_7	0	—	1	—	$\overline{AC}$
2	m_5, m_7, m_{13}, m_{15}	—	1	—	1	$\overline{BD}$
0	m_0, m_2	0	0	—	0	$\overline{A\overline{B}\overline{D}}$
1	m_2, m_{10}	—	0	1	0	$\overline{BCD}$
2	m_9, m_{13}	1	—	0	1	$\overline{ACD}$
2	m_{12}, m_{13}	1	1	0	—	$\overline{ABC}$

Step 4:

PI	Minterms	0	3	5	6	7	10	12	13	
$\overline{AC}$	m_3, m_6, m_7		⊗		⊗	×				✓
$\overline{BD}$	m_5, m_7, m_{13}			⊗		×			×	✓
$\overline{A\overline{B}\overline{D}}$	m_0	⊗								✓
$\overline{BCD}$	m_{10}						⊗			✓
$\overline{ABC}$	m_{12}, m_{13}							⊗	×	✓
$\overline{ACD}$	m_{13}								×	

$$F = \overline{AC} + \overline{BD} + \overline{A\overline{B}\overline{D}} + \overline{BCD} + \overline{ABC}$$

1.3 More Examples:

You can practice more examples by yourself or solve problems from the K-map exercises and check whether the answer matches with the K-map solution.

Try different combinations of minterms and don't care terms to get more familiar with the method. And also try to simplify expression with more variables (5 or 6) to see the advantage of this method over K-maps.

1.4 Tools:

Quine McCluskey Solver

For learning and practice, you should not use this during exams.

1.5 Tools:

[Handwritten Note on Quine McCluskey Solver](#)